

Plagiarism Report

File: 5f82415e648443a08d1e27bbc5351736.pdf

Plagiarism Percentage: 0.38%

Decentralized Certificate Generation and Verification using Ethereum Blockchain and IPFS: A Secure and Transparent Framework Aryan Raj Computer Science & Engineering Sharda University Greater Noida, India 2022006613.aryan@ug.sharda.ac.in Purushottam Choudhary Computer Science & Engineering Sharda University Greater Noida, India 2022592906.purushottam@ug.sharda.ac.in Mamta Narwaria Computer Science & Engineering Sharda University Greater Noida, India mamta.narwaria@sharda.ac.in

Abstract— Conventional digital certificate infrastructures, which are generally centralized, suffer from major issues such as forgery, tampering, and delays in verification procedures. This paper introduces a decentralized solution that uses the Ethereum blockchain and the Interplanetary File System (IPFS) to enhance certificate issuance and verification. Our system utilizes Ethereum smart contracts to produce automated, transparent, and secure certificate generation, while IPFS offers strong and decentralized storage for certificate files. Cryptographic methods provide secure user authentication and digital signature checking, thus precluding vulnerabilities that are inherent to centralized systems.

Additionally, the method proposed resolves scalability and optimization of transaction fees, thus proving to be an effective solution for academic, professional, and government use.

The implementation steps, system architecture description, and extensive security and performance analysis are all presented. Future plans involve the implementation of Layer-2 solutions and other advanced cryptographic techniques like zero-knowledge proofs to further promote privacy and efficiency.

I. INTRODUCTION The growing need for secure, transparent, and decentralized verification systems has given rise to tremendous research in blockchain-based solutions. In this paper, we design a system to generate and verify certificates with Ethereum and IPFS. With services like Pinata integrated for ease of IPFS file management, every certificate is uploaded to IPFS—resulting in a unique Content Identifier (CID) that is then stored on the Ethereum blockchain. This CID is a digital fingerprint; if there is any change in the certificate, it will change the CID, and tampering will be evident at once. The conventional certificate management based on centralized authorities is susceptible to security threats and inefficiencies. Our decentralized method, however, offers an immutable, verifiable record that increases trust and transparency. This study not only presents the technical framework but also discusses possible applications in education, healthcare, and industry, showing the system's capability to solve common problems like unauthorized duplication and data loss.

II. LITERATURE REVIEW

Decentralized certificate management has been made possible by recent research. Rahman et al. [1] proposed the Verifi-Chain approach, which makes use of blockchain and IPFS for secure digital certificate management. Oluwaseyi and Akinyede [2] also presented

a university certificate verification system based on blockchain that focuses on both scalability and security. Rahman et al. [3] subsequently took the field forward with a certificate authentication system that includes error correction, and Suganthalakshmi et al. [4] introduced a validation system that highlights the advantage of immutability. Xu [5] looked specifically at designing academic credential verification systems and featured enhanced user trust and data integrity. Sultana et al. [6] provided a conceptual framework utilizing IPFS and smart contracts to minimize certificate fraud, while Gangwar and Chaurasia [7] proposed QR code-based authentication to ease user verification. Further background is given by numerous Wikipedia articles describing Certificate of Authenticity [8], BLS digital signatures [9], OpenTimestamps [10], and Certificate Transparency [11]—each of which places the technical context in which decentralized systems exist. Performance evaluations by Pawar et al. [12] and security verifications by Prasan et al. [13] present quantitative proof of the superiority of the blockchain over centralized systems. Lastly, novel solutions like DocCert [14] and DClaims [15] demonstrate how decentralized storage and annotation mechanisms further augment document authenticity and censorship resistance.

III. SYSTEM ARCHITECTURE

The envisaged digital certification system is founded upon a layered and modular architecture that facilitates security, scalability, and usability. The architecture is structured into three major components:

1. **Blockchain Components:** This module acts as the system backbone, by handling certificate lifecycle via Ethereum smart contracts in Solidity. The contracts perform certificate issuing, verification, and revocation, and store metadata—in this case, the IPFS CID—on the immutable Ethereum block chain. Deployment and migration are handled through the Truffle framework, such that the contracts are compiled and deployed efficiently. This decentralized process greatly minimizes the risk of tampering and counterfeiting since any unauthorized modification is immediately traceable on the blockchain [1], [3].
2. **Application & User Interface:** The application layer serves as the middleman between the blockchain and end users. Implemented in Python using the Streamlit framework, it offers an interactive and user-friendly web-based interface. This layer accommodates multiple user roles such as institutions, verifiers, and regular users with specific functionalities. The backend utilizes Web3.py for interaction with the Ethereum network, and Firebase is used to integrate secure user authentication and role-based permissioning. Certificate generation, verification, and storage operations are handled by utility modules that make the system modular and scalable [7].
3. **Deployment and Development Tools:** Containerization using Docker makes consistent deployment over varied environments easier. Separate Dockerfiles for the blockchain network (with Ganache as an emulator) and the Streamlit application server are coordinated using Docker Compose. Truffle and Ganache are the foundation of the development environment, making smart contract compilation, migration, and testing easier. The containerized deployment approach reduces compatibility problems and makes updates easy, thus improving system reliability overall [2].

IV. IMPLEMENTATION DETAILS

Implementation of the decentralized certificate generation and validation system is designed to provide security, usability, and performance. The architecture of the system has three major modules: Blockchain Implementation, Application and User Interface Implementation, and Deployment and

Development Implementation. All of this work synergistically to implement a solid and dependable certificate management system utilizing blockchain and decentralized storage technologies.

1. Blockchain Implementation: The blockchain technology is the foundation of the system, providing secure and transparent issuance, verification, and revocation of certificates. The main smart contract, `Certification.sol`, is written in Solidity and deployed on the Ethereum blockchain using the Truffle framework. The smart contract is created to manage the whole certificate life cycle, such as the issuance, validation, and possible revocation of digital certificates. Every certificate that is issued by the system includes necessary properties like the ID of the certificate, the issuer information, the recipient information, the date of issuance, and a hash of the certificate data. The hash is recorded on the blockchain, and the original certificate file is stored in the Interplanetary File System (IPFS). The IPFS gives a Content Identifier (CID), which is used as the digital fingerprint of the file. This CID is kept in storage on the blockchain within the certificate record, meaning that any change to the certificate file leads to a different CID, and thus detecting tampering instantly. To maximize gas consumption and reduce transaction fees, the smart contract is coded with optimized data structures and algorithms. Intricate operations are decomposed into modular functions so that the contract only performs the required actions. This modularity minimizes computational overhead, making the system cost-efficient without compromising on security levels. Contract deployment and migration are automated with Truffle migration scripts. Automation guarantees that each version of the contract is deployed in a structured manner and thoroughly tested before production environment integration. Local testing and simulation are conducted using Ganache, a local Ethereum blockchain emulator, ensuring thorough testing without real transaction cost expenditure. Smart contract operations are:

1. Issue Certificate: Creates and adds a new certificate to the blockchain.
2. Revoke Certificate: Mentions a certificate revoked and changes the status.
3. Verify Certificate: Verifies the authenticity and validity of a specified certificate by comparing the CID stored with the generated hash from the original file.
4. GetCertificateDetails: Fetches the certificate metadata from the blockchain for verification and auditing.

The blockchain usage also includes the use of safe cryptographic methods for digitally signing certificates. The digital signatures are then verified during verification, which prevents the certificate from being tampered with or damaged since it was issued.

2. Application & User Interface Implementation The application layer is the interface between end users and the blockchain network. The system's backend is coded in Python, utilizing mainly the Web3.py library to interact with the Ethereum blockchain. The main backend script, `app.py`, processes main functions like issuing new certificates, validating existing ones, and user authentication. User interface is designed employing the Streamlit framework that generates an interactive, intuitive web-based application for end-users. Role-based access is supported through the interface such that institutions, verifiers, and general users may engage with the system as per their assigned role. UI components are comprised of:

- Certificate Issuance: Organizations are able to upload certificate information and initiate issuance. Certificate information is hashed and uploaded into IPFS, and the resultant CID is saved on the blockchain.
- Certificate Verification: Users and verifiers can input a certificate ID or upload a certificate file to confirm its genuineness. The

system takes the stored CID from the blockchain and cross-checks it with the uploaded file's hash to ensure integrity. • **Dashboard:** Gives a summary of issued, revoked, and validated certificates, and also includes functionality to handle user profiles and access history logs. • **Authentication and Authorization:** Integration with Firebase facilitates secure user authentication, where users can log in and control their profiles effectively. The backend communicates with the IPFS network using HTTP requests against Pinata, an IPFS cloud-based gateway. When a certificate is uploaded, it sends the file to IPFS using the Pinata API, which gives back a CID. The CID is thereafter stored on the blockchain as part of the record for the certificate. By integrating IPFS, certificate files are stored decentrally, minimizing dependency on a single point of failure and maximizing data availability. • **Flowchart on how Certificate is generated:-** • **Flowchart on how Certificate is verified:-** To enable effective processing and handling of data, utility functions are leveraged by the application. As an example, `cert_utils.py` is tasked with creating digital signatures and hashing certificate information, and `streamlit_utils.py` handles dynamic UI components and interactions. Through these utility functions, the complete certificate life cycle, from generation to verification, is streamlined.

3. Deployment & Development Implementation:

The deployment plan is optimized for maximum portability and scalability. The complete system is containerized through Docker, with the blockchain network and application server both encapsulated in isolated environments. There are individual Dockerfiles to create images for both the Ganache blockchain emulator and the Streamlit application server. Docker Compose is used to orchestrate the containers so that they can run simultaneously and communicate effectively. The steps involved in the deployment are:

1. **Building Docker Images:** Creating images for both the blockchain network and application server based on the corresponding Dockerfiles.
2. **Container Orchestration:** Employing Docker Compose to specify multi-container setups, with the blockchain emulator and application server being networked together.
3. **Environment Configuration:** Configuring environment variables to control API keys, blockchain network configurations, and database credentials.
4. **Continuous Integration/Continuous Deployment (CI/CD):** Automating the deployment process using CI/CD pipelines to minimize manual intervention and enhance reliability. The system is thoroughly tested with automated test scripts offered by the Truffle testing framework. The test scripts mimic normal certificate operations—issuance, verification, User institutions Access yes ✕ Certificate Generation ✕ Store Certification on IPFS Via Pinata ✕ Generate Unique CID (IPFS Content ID) Y 'STORES CID ON BLOCKCHAIN CERTIFICATE GENERATED AND STORED Verifier Access Retrieve CID From Blockchain ✕ Fetch Data From IPFS Using CID ✕ Compare Retrieved {CID & Blockchain CID} Certificate Verified Certificate invalid and revocation—to verify contract correctness and optimality in gas usage. Integration testing ensures seamless interaction between the backend API and blockchain and the correctness of the entire workflow.
5. **SECURITY & PERFORMANCE ANALYSIS:** Blockchain technology naturally provides data immutability and openness. In the system, every issued certificate is stored securely in the Ethereum blockchain, where it is cryptographically signed and anchored with a CID from IPFS. This makes it much less likely to be forged or illegally altered. The employment of digital signatures also boosts security, because the verification procedure

ensures that certificates issued by authentic entities are deemed valid. The decentralized architecture of the blockchain protects it from tampering and manipulation of data.

Furthermore, making any change in a certificate file on IPFS would create a new CID, which would make it uncorrelatable with the stored record in the blockchain. This implicit detection of tampering ensures certificate integrity during the lifetime of the certificates. 1.

Preventing Certificate Tampering & Forgery: The system uses robust cryptographic algorithms, including SHA-256 hashing and ECDSA digital signatures, to secure certificate data. Upon issuance, each certificate is hashed, and this hash is combined with the issuer's private key to generate a digital signature. The resulting signature is then stored on the blockchain alongside the CID. Verification involves recalculating the certificate's hash, comparing it with the blockchain record, and validating the digital signature. This layered approach ensures that no unauthorized changes can go undetected. 2.

Cost Optimazation for Large Scale Optimization: Gas expenses are a top priority for blockchain use cases. To keep transaction fees as low as possible, the smart contracts are engineered for performance. Unnecessary calculations are avoided, and intricate operations are decomposed into smaller gas-costing functions. Performance tests that were carried out using Ganache show that the cost of transactions is about 30,000 gas units per issuance, which is roughly \$0.50 per transaction. This optimization is sufficient to make the system scalable for big deployments while keeping it financially sustainable. 3.

Performance Benchmarking: Benchmarking was done to compare transaction latency, gas usage, and certificate retrieval speeds. The average latency for certificate issuance was around 2.5 seconds, while retrieval from IPFS was about 1.2 seconds. While a bit slower than in centralized setups, this performance comes at the cost of enhanced security and transparency. The gas optimization techniques built into the system ensure cost-

effectiveness even under heavy usage. 4. **Comparison:** In comparison to centralized verification systems, blockchain-based methods prove to be more secure and sound in data. Centralized systems can issue and verify certificates in the blink of an eye but are susceptible to tampering and data theft. However, blockchain systems, at additional expense and latency, offer an immutable record that is resistant to manipulation and unauthorized changes. This speed-security trade-off is worthwhile in environments where data integrity is essential. VI. CHALLENGES & LIMITATIONS: 1.

Scalability Concerns: One of the key challenges is to scale the system to accommodate more transactions. When the network is congested, fees on Ethereum can increase dramatically, potentially affecting the usability of the system for mainstream adoption. The scalability question requires ongoing innovation and perhaps even exploring other or additional blockchain platforms to preserve efficiency as usage increases. 2. **Challenges For Non-Technical Users:** Even with attempts to create an easy-to-use user interface, the inherent technological complexity of blockchain and decentralized storage systems might still create challenges for non-technical users.

Making user interactions easy and detailed guidelines are crucial in order to make the advantages of the system reach as many people as possible, making it easier to adapt to various user groups. 3. **Potential Privacy Issues:** Although IPFS provides decentralized and fault-tolerant storage, privacy issues are raised as well. Sensitive certificate data needs to

be properly controlled in order not to be accessed without authority, as IPFS-stored data is

openly accessible unless it has been encrypted suitably. Strong encryption methodologies and secure access controls are essential in order to protect privacy while taking advantage of decentralized storage.

4. Integration Challenges: Integrating the decentralized system with existing centralized verification procedures is not without challenges. There are technical and regulatory issues in making sure that both systems would work together harmoniously. Creating standardized APIs and middleware solutions will be important in opening the gap between legacy and decentralized systems, providing seamless interoperability as well as consistency in data.

VII. FUTURE DIRECTIONS

1. Layer-2 Solutions: To address scalability challenges and elevated transaction fees, future development should seek to incorporate Layer-2 solutions—like Polygon and Optimistic Rollups—that enable transactions to be executed off the Ethereum mainnet but retain the security benefits of the same. Those solutions have the potential to keep costs low and enhance throughput without compromising on the system being decentralized.

2. Zero-Knowledge Proofs: More advanced cryptographic methods such as zero-knowledge proofs (ZKPs) hold potential for boosting privacy. Through verification of the authenticity of certificates without revealing sensitive underlying data, ZKPs can strengthen the security framework of the system and safeguard user data more securely.

3. Cross-chain Interoperability: Increasing interoperability between several blockchain networks is another area of future research. By enabling the certification system to communicate smoothly with various blockchain platforms, it would be feasible to take advantage of the strengths of each network, thereby enhancing scalability, robustness, and user choice.

4. Legal and Regulatory Considerations: As decentralized systems grow in popularity, legal and regulatory issues must be addressed. The future research must be aimed at creating frameworks that are compliant with regional and global regulations so that decentralized certification may be adopted safely and legally within different jurisdictions.

5. AI Integration: Adding artificial intelligence to the system may offer a secondary level of security in the form of real-time fraud detection. Machine learning can be used to scan transaction patterns and identify irregularities in certificate issuance, allowing for proactive steps against potential attacks and further improving system reliability.

VIII. CONCLUSION

The decentralized certificate creation and validation system outlined in this paper utilizes the Ethereum blockchain and IPFS to counter the shortcomings of conventional centralized certificate management. Utilizing the transparency and immutability of blockchain technology and combining it with distributed storage, the system attains secure, tamper-evident, and efficient certificate management. While the blockchain-based solution presents some issues, including increased transaction cost and latency relative to centralized systems, its advantages regarding security, trust, and robustness are noteworthy. In addition, the optimized smart contract semantics and modular app architecture render the solution scalable and flexible for various applications in academia, industry, and government. Future advancements—specifically, those in the realms of Layer-2 integration, sophisticated cryptographic privacy features, cross-chain bridging, and AI-driven fraud detection—hold the potential to enhance the system's performance and user experience even further. Overall, this research provides a strong foundation upon which the future generation of digital certification practices shall be built, in which data

integrity and trust are the highest priority. IX. REFERENCES [1] T. Rahman, S. I. Mouno, A. M. R. Raatul, A. K. A. Azad, and N. Mansoor, "Verifi-Chain: A Credentials Verifier using Blockchain and IPFS," arXiv preprint arXiv:2307.05797, 2023. [Online]. Available: <https://arxiv.org/abs/2307.05797> [2] O. S. Oluwaseyi and R. O. Akinyede, "Utilizing Blockchain Technology for University Certificate Verification System," *Int. J. Appl. Inf. Syst.*, vol. 12, no. 45, 2024. [Online]. Available: <https://www.ijais.org/archives/volume12/number45/utilizing-blockchain-technology-for-university-certificate-verification-system/> [3] M. Rahman, M. Tonmoy, S. Shihab, and R. Farhana, "Blockchain-Based Certificate Authentication System with Enabling Correction," *J. Comput. Commun.*, vol. 11, pp. 73–82, 2023, Doi: 10.4236/jcc.2023.113006. [4] R. Suganthalakshmi, G. C. Praba, K. Abhirami, and S. Puvaneswari, "Blockchain-Based Certificate Validation System," *Int. Res. J. Mod. Eng. Technol. Sci.*, 2022. [Online]. Available: https://www.irjmets.com/uploadedfiles/paper/issue_7_july_2022/28889/final/fin_irjmets1659003745.pdf [5] Y. Xu, "Development of Blockchain-Based Academic Credential Verification System," *Open Access Libr. J.*, vol. 11, no. 9, 2024, Doi: 10.4236/oalib.1112130. [6] S. A. Sultana, C. Rupa, R. P. Malleswari, and T. R. Gadekallu, "IPFS-Blockchain Smart Contracts Based Conceptual Framework to Reduce Certificate Frauds in the Academic Field," *Information*, vol. 14, no. 8, p. 446, 2023, Doi: 10.3390/info14080446. [7] S. Gangwar and A. Chaurasia, "Blockchain-based Authentication and Verification System for Academic Certificate using QR Code and Decentralized Applications," *Int. J. Comput. Appl.*, vol. 186, no. 26, 2024, Doi: 10.1109/icscan53069.2021.9526377. [8] "Certificate of Authenticity – Non-Fungible Tokens as Digital Certificates," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Certificate_of_authenticity (accessed March 2025). [9] "BLS Digital Signature – An Overview," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/BLS_digital_signature (accessed March 2025). [10] "OpenTimestamps: Scalable, Trust-Minimized, Distributed Timestamping with Bitcoin," Wikipedia. [Online]. Available: <https://opentimestamps.org> (accessed March 2025). [11] "Certificate Transparency," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Certificate_Transparency (accessed March 2025). [12] M. K. Pawar et al., "Performance Analysis of E-certificate Generation and Verification Using Blockchain and IPFS," *Int. J. Comput. Res. Eng. Stud.*, 2022. [Online]. Available: https://www.psvpec.in/jcres/2022_2/87.pdf [13] U. D. Prasan et al., "Validating and Enhancing Security of Certificates Using the Ethereum Blockchain and IPFS," *Int. J. Eng. Res. Appl.*, vol. 13, no. 4, 2023, Doi: 10.9790/9622-13042531. [14] M. Aldwairi, M. Badra, and R. Borghol, "DocCert: Nostrification, Document Verification and Authenticity Blockchain Solution," arXiv preprint arXiv:2310.09136, 2023. [Online]. Available: <https://arxiv.org/abs/2310.09136> [15] J. Santos, N. Santos, and D. Dias, "DCClaims: A Censorship Resistant Web Annotations System using IPFS and Ethereum," arXiv preprint arXiv:1912.03388, 2019. [Online]. Available: <https://arxiv.org/abs/1912.03388> .